DI40001 DIICSU Honours Project

Structural design and mechanical simulation of vertical roll-bending equipment for thin-walled rings

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Abstract

This project presents the structural design and mechanical simulation of a vertical three-roll bending system intended for the continuous, integrated manufacture of large-diameter thin-walled stainless steel rings (target diameter ϕ12 m). Conventional horizontal roll bending equipment suffers from gravitational deflection of the workpiece, positional drift, and limited process integration when handling 4–10 mm semi-hard 304 stainless steel strip. To address these limitations, a vertical layout is adopted, coupling an uncoiling unit, a symmetric three-roll bending station, and a laser cutting module into a single automated line. The uncoiling device employs an expandable mandrel with hydraulic clamping to provide stable coil feeding, while hydraulic actuators on the working rolls enable precise displacement control throughout the bending stroke. The laser cutting unit uses a screw–slider drive and optimised power parameters to achieve cut quality compatible with downstream welding requirements. ANSYS/LS-DYNA nonlinear finite element analysis is used to simulate the roll bending process, capturing equivalent stress distributions and maximum deformation at successive forming stages. The influence of rear-roll centre distance on springback and dimensional accuracy is systematically investigated, leading to the selection of optimal process parameters. A subsequent ANSYS structural optimisation study reduces roll diameter and effective length while satisfying strength and stiffness constraints, yielding a lightweight design with lower manufacturing cost. The resulting equipment architecture achieves fully automated processing from coil to finished ring, with demonstrated improvements in forming accuracy and production efficiency.

Keywords: thin-walled ring parts; roll bending forming; structural design; mechanical simulation; finite element analysis; 304 stainless steel

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1 Introduction

1.1 Background and Motivation

Thin-walled ring components — generally defined as annular parts whose wall-thickness-to-inner-radius ratio is less than 1:20 — constitute critical load-bearing structures in a broad range of high-technology industries [1]. In the offshore wind energy sector, each turbine tower requires numerous large cylindrical shell rings: transition pieces (ϕ5–8 m, wall 50–80 mm), monopile foundation sections (ϕ6–12 m, wall 60–100 mm), and internal stiffening rings (ϕ3–12 m, wall 4–20 mm) that sit within the capability range of the equipment designed in this project [2]. In the chemical and process industry, pressure vessel shells manufactured to ASME VIII or EN 13445 demand circularity tolerances of $\Delta R/R \leq 1\%$ and weld-joint radial offsets below $0.25t$ [3]. In the aerospace sector, propellant tank skirt rings, access manhole flanges, and heavy-launch-vehicle body rings reaching ϕ8–12 m must simultaneously withstand extreme fatigue, thermal gradients, and corrosive environments, placing exceptional demands on both material selection and forming precision [1].

Among candidate materials for these applications, austenitic 304 stainless steel occupies a dominant position owing to its combination of yield strength ($\sigma_y \approx 205$ MPa), fracture elongation ($\approx 45\%$), and corrosion resistance [4]. However, its high work-hardening rate and pronounced springback behaviour make the continuous roll bending of 4–10 mm semi-hard strip into rings of diameter ϕ12 m a technically demanding problem. Conventional horizontal roll bending machines are susceptible to gravitational sag of the workpiece under self-weight, which introduces out-of-plane distortion and degrades ring circularity [5]. Moreover, traditional multi-stage workflows — in which uncoiling, bending, and trimming are performed as separate operations on independent machines — accumulate positioning errors at each inter-machine transfer and yield low overall throughput.

A vertical roll bending layout, in which the bending plane is perpendicular to the gravitational vector, eliminates the dominant self-weight deflection mechanism and provides a natural basis for integrating coil feeding, roll bending, and laser cutting into a single automated production line. Su et al. [3] demonstrated a 40% floor-space reduction and a 25% circularity improvement for a vertical stainless-steel ring forming line over its horizontal equivalent, attributing the improvement almost entirely to the elimination of gravitational sag. Despite these documented advantages, published structural design studies and simulation data for large-scale vertical roll bending systems targeting ϕ12 m rings from 4–10 mm 304 stainless steel remain absent from the open literature, motivating the present investigation.

1.2 Roll Bending Technology: Principles, Configurations, and Challenges

1.2.1 The Roll Bending Process and Its Advantages

Roll bending — also referred to as plate rolling or cold roll forming — is a continuous three-point bending process in which flat sheet metal is passed between a set of

cylindrical rolls whose relative positions define the curvature imparted to the workpiece [6]. Unlike press bending or stretch forming, roll bending achieves the target curvature through incremental plastic deformation accumulated over multiple passes, making it particularly suited to large-radius, large-diameter ring components from wide strip material [7]. The process offers several well-documented advantages: minimal tooling investment (no dedicated die required), high material utilisation, rapid changeover between product diameters by roll-position adjustment, and compatibility with coil-fed automation [5]. For these reasons, roll bending is the dominant process for fabricating large cylindrical shells in the offshore energy, pressure vessel, and aerospace sectors.

The sheet undergoes three distinct deformation regimes as bending progresses. In the elastic regime the cross-section deforms reversibly and the strip returns flat on unloading. Once the outer fibres yield, the elasto-plastic regime is entered: an elastic core persists at the neutral axis while the outer fibres follow the plastic flow rule. At sufficiently large displacements the fully plastic regime is reached. For the bending ratio $R/t \approx 600$ applicable to the present design, the elasto-plastic regime prevails, making accurate springback prediction essential [6]. Springback — the elastic recovery of curvature upon removal of the applied moment — is approximately 60% larger in 304 stainless steel than in mild steel at the same bending ratio [4], underscoring the need for carefully optimised process parameters.

Common forming defects in roll bending of thin-walled sheets include: springback-induced radius deviation; longitudinal edge waving from compressive buckling of strip edges; transverse bowing from non-uniform longitudinal strain across the width; wall thickness non-uniformity from differential fibre elongation; and surface marking from inadequate roll hardness [6]. For the wide thin strips in this project (width up to 2200 mm, thickness as low as 4 mm), edge waving is a particular risk, necessitating a superimposed strip tension of 2–5 kN to suppress compressive edge stress throughout the bending cycle.

1.2.2 Equipment Configurations

Commercial roll bending machines are classified by the number and arrangement of their working rolls [5]. The principal configurations are described below, with key characteristics summarised in Table 1.

Two-roll configuration. A two-roll machine uses one rigid and one elastic (polyurethane-wrapped) roll [8]. The process is simple and fast, but each product diameter requires a dedicated rigid roll, making two-roll machines unsuitable for multi-diameter production.

Three-roll symmetric configuration. The most widely used configuration for large-ring production [9]. Two bottom support rolls are symmetrically disposed about the vertical centreline of the fixed top drive roll. Strip curvature is controlled by vertical displacement of the bottom rolls. The symmetric layout produces balanced lateral forces at the machine frame and is readily adapted to a vertical orientation, as shown in Figure **??**. Its principal limitation is the existence of an unbent length at each strip end that must be pre-bent separately.

Three-roll asymmetric configuration. One bottom roll is replaced by a laterally adjustable side roll, enabling pre-bending of one strip end without repositioning. However, the asymmetric force resultant increases the required frame section modulus by

approximately 35% [3], and vertical integration of this layout is mechanically complicated.

Four-roll configuration. A fixed top roll, a vertically adjustable bottom roll, and two side rolls provide both pre-bending and main bending capability in a single setup [5]. Four-roll machines offer excellent diameter flexibility and end-straightening, but their structural complexity, high capital cost, and difficulty adapting to a vertical orientation make them unsuitable for the present application.

For this project, the three-roll symmetric vertical configuration was selected based on its structural simplicity, balanced loading, low capital cost, and inherent compatibility with vertical layout integration.

1.2.3 Research Status and Technology Gap

The theoretical analysis of roll bending has developed from simplified analytical methods dating to the 1960s — in which transverse bending and longitudinal deformation were treated separately using pure-bending theory [6] — to sophisticated finite element (FE) simulation approaches. Early analytical models (Nakajima and Sato [9]; Walker and Pick [7]) established the geometric relationship between roll positions and imparted curvature. Subsequent FE studies by Liu et al. [10], Zhang et al. [11], and Zeng et al. [8] demonstrated that combined isotropic-kinematic hardening models reduce springback prediction error for 304 stainless steel to below 5%, while rear-roll centre distance emerges as the most influential geometric process parameter governing wall thinning and radius deviation.

In terms of industrial equipment, the high-end large-ring roll bending market is dominated by European manufacturers — Haeusler (Switzerland), Roundo (Sweden), DAVI (Italy), and Hugh Smith (UK) — whose machines are predominantly horizontal in layout and rely on operator skill for iterative circularity correction [5]. Domestic Chinese manufacturers (Taian Hualv, Nanjing Huanli Heavy Industry, Shenyang Heavy Machinery) have achieved mature production technology but remain behind international leaders in bending accuracy, automation level, and large-format capability. Vertical configurations have received comparatively little attention: Su et al. [3] reported a vertical stainless-steel ring forming line achieving 25% circularity improvement over a horizontal equivalent, but provided no detailed structural analysis of the roll and frame assembly. No published study combines a vertical three-roll layout, 304 stainless steel at $\phi 12\,\mathrm{m}$ target diameter, and in-line laser cutting within a single validated design and simulation framework.

1.3 Project Aims, Objectives, and Scope

The overarching aim of this project is to produce a structurally validated design and a finite element simulation framework for an integrated vertical roll bending production line capable of manufacturing $\phi 12\,\mathrm{m}$ thin-walled 304 stainless steel rings to industrial circularity standards ($\Delta R/R < 1\%$). Five specific objectives support this aim: (i) a systematic comparative evaluation of roll bending configurations against the technical requirements of the target product; (ii) detailed SolidWorks CAD models of all three sub-assemblies with kinematic compatibility verified; (iii) a validated ANSYS/LS-DYNA nonlinear FE model of the roll bending process, used to quantify stress, deformation, and springback as functions of rear-roll centre distance; (iv) a parametric structural optimisation of the

top drive roll using ANSYS Design Exploration; and (v) evaluation of the overall system against design requirements and comparison with published benchmarks.

The project scope encompasses the structural and mechanical design of the forming equipment and the FE simulation of the roll bending process. The following are explicitly outside scope: detailed electrical and hydraulic control system design; thermal analysis of the laser cutting zone; welding process simulation; and physical prototype construction or experimental validation on a full-scale machine. These limitations are discussed further in Section 7.

2 Literature Review

2.1 Thin-Walled Ring Components: Applications and Manufacturing Demands

Thin-walled ring components are widely employed across high-value industries as primary load-bearing structural elements. In the energy sector, wind turbine towers and offshore platform monopiles require large cylindrical shell rings; in the chemical and nuclear industries, pressure vessels and reactor containment shells demand rings of strictly controlled circularity ($\Delta R/R \leq 1\%$ per ASME VIII UG-80) and weld-joint alignment [3]. In the aerospace sector, rocket body segments, propellant tank skirts, and turbine casing rings must simultaneously satisfy tight dimensional tolerances and withstand extreme mechanical and thermal loading [1], [2]. These components are trending towards larger diameter, thinner walls, lighter mass, and higher reliability, placing increasingly stringent demands on the forming process [1].

Among the candidate forming routes — press bending, stretch bending, creep-age forming, shot peening, and roll bending — roll bending occupies a dominant position for large-diameter thin-walled cylindrical rings owing to its low tooling investment, high material utilisation, rapid diameter changeover, and compatibility with coil-fed automation [5]. Press bending suffers from long cycle times and low throughput for large rings; creep-age forming and shot peening involve high capital cost and are primarily suited to aluminium alloys rather than stainless steels. Roll bending is therefore the process of choice for the $\phi 12\,\mathrm{m}$ 304 stainless steel rings targeted in this project.

2.2 Existing Roll Bending Equipment and Its Limitations

Conventional roll bending machines are predominantly horizontal in layout, with the roll axes oriented horizontally and the bending plane vertical. Leading international manufacturers — Haeusler (Switzerland), DAVI (Italy), Roundo (Sweden), and Hugh Smith (UK) — produce high-end horizontal plate bending machines that have set the industry benchmark for large-ring production [5]. Domestic Chinese manufacturers, including Taian Hualv, Nanjing Huanli Heavy Industry, and Shenyang Heavy Machinery, have achieved progressively improved CNC roll bending capability but remain behind international leaders in forming accuracy and large-format automation.

Horizontal machines suffer from a fundamental physical limitation when processing large-diameter thin-walled strips: the workpiece sags under its own self-weight in the unsupported span between rolls. For a 304 stainless steel strip of width 2200 mm and thickness 4 mm, the mid-span gravitational deflection in a horizontal machine is estimated at 12–18 mm, which directly translates into an out-of-plane circularity error in the finished ring [5]. Wang et al. [5] measured residual ovality of 0.3–0.8% after a three-pass bending sequence on a state-of-the-art CNC horizontal machine for 3–6 mm mild steel shells, confirming that gravitational sag is the dominant source of circularity error even with modern CNC correction.

Beyond the self-weight issue, traditional horizontal production lines perform uncoiling, roll bending, and cutting as separate operations on independent machines. Each inter-machine transfer introduces positioning errors and handling time, reducing overall throughput and increasing the risk of workpiece damage. For the large, flexible strips considered here (thickness 4–10 mm, length up to 38 m per ring), this multi-stage workflow is particularly problematic: the strip must be re-aligned at each station, and cumulative errors degrade the final ring quality.

2.3 Vertical Roll Bending: Emerging Solutions and Design Challenges

A vertical roll bending layout, in which the roll axes are oriented vertically and the bending plane is horizontal, eliminates the self-weight deflection mechanism entirely: the strip lies in the bending plane and gravity acts perpendicular to the forming direction [3]. Su et al. [3] reported a 25% improvement in ring circularity and a 40% reduction in floor space for a vertical stainless-steel ring forming line compared with a horizontal equivalent, validating the theoretical advantage of the vertical layout.

The vertical configuration also enables process integration: the uncoiling unit, roll bending station, and cutting module can be arranged in a vertical in-line sequence, eliminating inter-machine transfers and enabling fully automated processing from coil to finished ring [3]. This integration reduces cumulative positioning errors, decreases floor space, and improves production throughput — all critical for the small-batch, high-precision production of ϕ12 m rings.

Despite these advantages, the design of large-scale vertical roll bending equipment for ϕ12 m rings from 4–10 mm 304 stainless steel presents several unresolved engineering challenges. The peak forming forces and the large roll span (up to 2200 mm) impose demanding structural requirements on the roll and frame assembly that have not been addressed in the published literature for this scale of machine. The springback behaviour of semi-hard 304 stainless steel at large bending radii ($R/t \approx 600$) under the specific geometric constraints of a vertical three-roll machine has not been characterised, making it impossible to select rear-roll centre distance and displacement setpoints from existing data alone. Finally, the structural integrity and mass of the working rolls under distributed forming load must be rigorously optimised to ensure elastic deflection does not degrade ring circularity [12].

2.4 In-Line Laser Cutting for Integrated Ring Production

Integrating a cutting step within the roll bending line requires a technology that imposes no contact forces on the as-formed ring, achieves positioning accuracy compatible with welding joint preparation requirements, and operates at a speed consistent with the bending cycle time. Fibre laser cutting satisfies all three requirements: contact-force-free operation eliminates distortion risk, positioning repeatability of ± 0.1 mm is achievable with servo-driven stages, and cutting speeds of 1.0–2.5 m/min for 4–10 mm 304 stainless steel are consistent with bending line speed [13].

Wu et al. [13] established parameter windows for fibre laser processing of cold-rolled 304 stainless steel compatible with AWS D1.6 TIG butt weld preparation: laser power 4.0–5.5 kW, cutting speed 1.0–2.5 m/min, nitrogen assist gas pressure 10–18 bar, and focal position 1–2 mm below the top surface. Within these windows, HAZ width remained below 0.28 mm, cut-face roughness R_a below 3.2 μm, and dross height below 0.1 mm. Fibre lasers (1070 nm) are preferred over CO_2 lasers (10.6 μm) because the shorter wavelength is absorbed at approximately 35% efficiency by 304 stainless steel versus approximately 8% for CO_2, enabling 30% lower specific energy consumption at equivalent cut quality [13].

2.5 Research Gap and Project Motivation

The foregoing review identifies a clear and unaddressed engineering gap. Existing horizontal roll bending equipment is fundamentally limited by gravitational self-weight deflection for large thin-walled strips and relies on multi-stage workflows that accumulate positioning errors. Vertical roll bending configurations have been demonstrated at smaller scales [3] but no published structural design study addresses a vertical machine targeting ϕ12 m rings from 4–10 mm 304 stainless steel with integrated laser cutting. The springback characteristics of 304 stainless steel at this scale, the structural integrity of the working rolls under peak forming loads, and the optimised roll geometry that minimises mass while satisfying stiffness and strength constraints all remain uncharacterised in the literature.

This project directly addresses these gaps by presenting a fully integrated vertical three-roll bending system design for ϕ12 m rings, ANSYS/LS-DYNA nonlinear FE simulation to determine optimal process parameters and springback corrections, and ANSYS Design Exploration structural optimisation of the working rolls. Together, these contributions provide a validated equipment design and simulation framework that advances the state of the art in large-diameter thin-walled ring production.

3 System Design

3.1 Overall Design Scheme

3.1.1 Overall Layout and Functional Requirements

The target product is a seam-welded cylindrical ring of nominal bore diameter ϕ12 m, formed from cold-rolled 304 stainless steel coil strip in the thickness range 4–10 mm

and width range 800–2200 mm. The finished ring must satisfy a circularity tolerance of $\Delta R/R \leq 1\%$, consistent with ASME VIII Division 1 requirements for butt-welded pressure vessel shells [3].

Since the raw material arrives at the workshop in the form of coil, and to improve manufacturing accuracy and efficiency while reducing the self-weight deformation of large thin-walled rings, all three process steps — vertical uncoiling, vertical roll bending, and cutting — are performed at a single station. The complete integrated system comprises four principal sub-assemblies: (i) a base platform that supports the formed ring during and after bending; (ii) a vertical expandable-mandrel uncoiling unit; (iii) a symmetric three-roll vertical bending station; and (iv) a laser cutting module. The overall system occupies a floor footprint of 146 m $\times$ 14.6 m with a total assembly height of 5630 mm, forming a complete large thin-walled ring manufacturing cell. The overall equipment layout is shown in Figure 1.

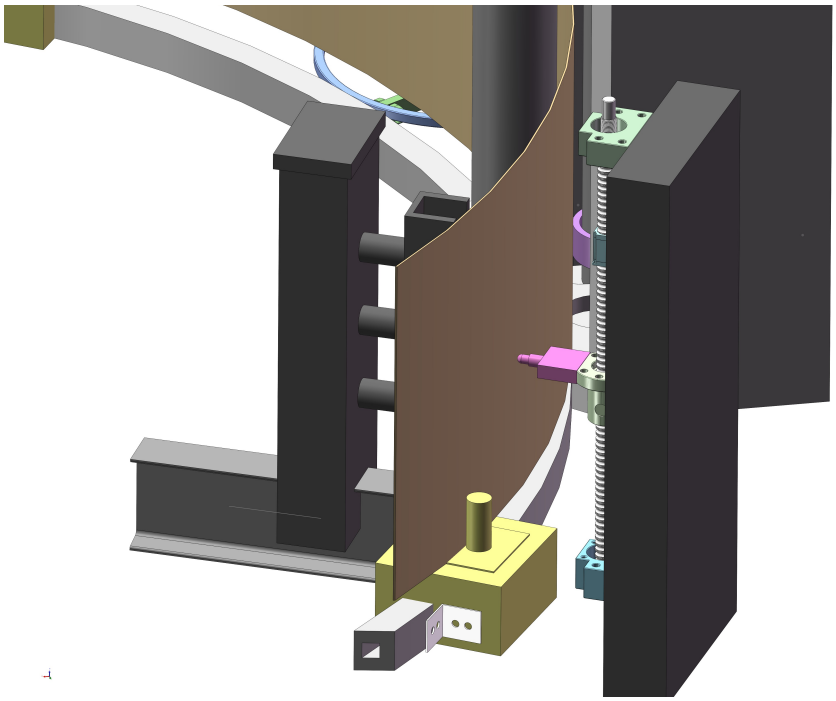


Figure 1: Overall layout of the integrated vertical roll bending production system. From left to right: vertical uncoiling unit, three-roll bending station, laser cutting module, and base platform.

3.1.2 Configuration Selection

Based on the working principles of two-roll, three-roll, and four-roll roll bending configurations reviewed in Section 2, a systematic comparative evaluation was conducted against the key selection criteria for this application, as summarised in Table 1.

Table 1: Comparative evaluation of roll bending configurations

Criterion	**2-Roll**	**3-Roll Sym.**	**3-Roll Asym.**	**4-Roll**
Diameter flexibility	Poor	Good	Good	Excellent
End-straightening	Poor	Moderate	Good	Excellent
Structural complexity	Low	Low	Medium	High
Capital cost	Medium	Low	Medium	High
Vertical integration	Feasible	Feasible	Difficult	Impractical
Force balance (vertical)	Unbalanced	Balanced	Unbalanced	Balanced
Maintenance access	Good	Good	Moderate	Poor
Overall rating	Poor	**Best**	Moderate	Moderate

The four-roll configuration, while capable of pre-bending strip ends, is rejected due to its high structural complexity and the near-impossibility of integrating its four-point force system into a compact vertical layout [5]. The three-roll asymmetric variant is rejected because its unbalanced lateral frame reaction increases the required frame section modulus by approximately 35% and requires manual strip repositioning for opposite-end pre-bending, reducing productivity [3]. The two-roll configuration requires a dedicated rigid roll for each product diameter, which is economically prohibitive for the product mix targeted.

The symmetric three-roll vertical configuration is therefore adopted as the design scheme. The top drive roll (Axis 1) is driven by a hydraulic motor and performs rotational motion to feed the strip and apply bending torque. The two bottom support rolls (Axes 2 and 3) are driven by hydraulic cylinders along inclined linear guides to provide the pressing displacement that controls strip curvature. By varying the relative position of the bottom rolls with respect to the top roll and the rotational speed of the top roll, different degrees of bending can be imparted to satisfy different target curvature radii [6]. This configuration has the advantages of simple structure, light weight, and ease of manufacture, and is well suited to the thin-sheet material (4–10 mm) targeted in this project. The vertical orientation eliminates gravitational self-weight deflection of the strip and allows direct vertical integration of the uncoiling unit above the bending station, avoiding multi-stage material transfer.

3.2 Working Roll Design

3.2.1 Forming Force Analysis

Accurate estimation of the peak forming force is the critical first step in sizing the working rolls, hydraulic drive cylinders, and main frame. The roll bending process is modelled as symmetric three-point bending of a wide strip. For strip thickness t, width b, and bottom-roll centre half-distance $a = d_{13}/2$, the peak bending moment at the top roll under full-width plastic bending is [6]:

$$M_b = \frac{\sigma_y b t^2}{4} \tag{1}$$

where $\sigma_y = 205\,\text{MPa}$ is the 0.2% proof stress of semi-hard 304 stainless steel [4]. For the maximum-load case ($t = 10\,\text{mm}$, $b = 2200\,\text{mm}$) and $a = 1100\,\text{mm}$:

$$F = \frac{2M_b}{a} = \frac{\sigma_y b t^2}{2a} = \frac{205 \times 2200 \times 10^2}{2 \times 1100} = 205{,}000\,\text{N} \approx 205\,\text{kN} \tag{2}$$

A safety factor of 1.5 is applied to account for dynamic effects, friction at the roll–strip contact interface ($\mu \approx 0.12$), and strip work hardening during the first bending pass, yielding a design forming force of $F_d = 308\,\text{kN}$ per bottom roll. This value governs both the hydraulic cylinder bore selection and the structural sizing of the roll journals.

3.2.2 Top Drive Roll (Front Roll)

The top roll (Axis 1) is the primary drive member. It is driven in rotation by a hydraulic motor and must transmit the full bending torque to the strip surface while sustaining the combined reaction force from both bottom rolls. Based on the force analysis above, the top roll journal must withstand a total downward reaction of $2F_d = 616\,\text{kN}$. The roll body is manufactured from 42CrMo4 alloy steel (quenched and tempered, hardness 280–320 HB) with a nominal barrel diameter $D_1 = 500\,\text{mm}$ and effective roll length $L_1 = 2600\,\text{mm}$, consistent with the maximum product width of 2200 mm plus 200 mm over-run on each side. The roll body diameter of 500 mm provides a cross-sectional second moment of area $I = \pi D_1^4/64 = 3.07 \times 10^9\,\text{mm}^4$, yielding a mid-span elastic deflection under the peak distributed forming load of [12]:

$$\delta = \frac{5 w L_1^4}{384 E I} = \frac{5 \times 118.5 \times 2600^4}{384 \times 210{,}000 \times 3.07 \times 10^9} \approx 0.62\,\text{mm} \tag{3}$$

where $w = 2F_d/L_1 = 118.5\,\text{N/mm}$ is the distributed forming load intensity and $E = 210{,}000\,\text{MPa}$ is the Young's modulus of 42CrMo4. The maximum deflection of 0.62 mm equates to 0.24 mm/m, satisfying the design requirement of $\leq 0.25\,\text{mm/m}$ for ring circularity.

The roll surface is hard-chrome plated to a depth of 0.12 mm and finish-ground to a surface roughness of $R_a \leq 0.8\,\mu\text{m}$. Both roll journals are supported in self-aligning spherical roller bearings housed in the upper cross-beam of the main frame, allowing minor angular misalignment caused by roll deflection without inducing edge loads on the bearing raceways.

3.2.3 Bottom Support Rolls (Rear Rolls)

The two bottom rolls (Axes 2 and 3) are geometrically identical and are arranged symmetrically about the vertical centreline of the top roll. Each roll has nominal barrel diameter $D_2 = 450\,\text{mm}$, effective roll length $L_2 = 2600\,\text{mm}$, and is manufactured from 42CrMo4 (quenched and tempered, 280–320 HB). The smaller diameter relative to the top roll is acceptable because each bottom roll carries only one-half of the total forming force reaction. The bottom roll journals are mounted in linear sliding bearing housings constrained to inclined guide channels at 45° to the vertical, enabling controlled simulta-

neous displacement of both bottom rolls towards or away from the top roll centreline. The centre half-distance between the bottom roll axes and the top roll axis, $a = d_{13}/2$, is the primary process parameter governing imparted curvature and is adjusted by the hydraulic cylinder displacement setpoints derived from the FE parametric study (Section 5).

3.3 Hydraulic Drive System

All primary motions of the bending station — top-roll rotation and bottom-roll translation — are actuated hydraulically. Hydraulic drive is preferred over electromechanical alternatives for this scale of machine because it offers high power density, inherent overload protection through pressure relief valves, and continuous force control throughout the stroke [5].

The top roll rotational drive uses a low-speed high-torque hydraulic motor (displacement 5000 cm^3/rev) coupled directly to the roll journal via a rigid flanged coupling. The motor is powered from a variable-displacement axial-piston pump (nominal flow 200 L/min, maximum pressure 25 MPa). The drive torque required to overcome strip bending resistance and rolling friction at maximum load is estimated at $T_{drive} \approx M_b + F_d \mu D_1/2 \approx 98\,\text{kN·m}$, which is within the rated output of the selected motor at 20 MPa operating pressure.

The bottom-roll displacement is controlled by two hydraulic cylinders (bore ϕ200 mm, stroke 600 mm, rated at 25 MPa), one per bottom roll. The cylinder rod is pin-jointed to the sliding bearing housing of each bottom roll, and the cylinder barrel is pin-jointed to the main frame, so that as the cylinder extends the bottom roll moves along its 45° guide channel. A servo-proportional directional control valve (Bosch Rexroth 4WRKE series, ±10 V command signal, response time <8 ms) on each cylinder enables closed-loop position control with 1 μm magnetostrictive displacement transducer feedback. The PLC position loop achieves a roll-centre displacement repeatability of ±0.05 mm (2σ), which translates directly to a radius repeatability of better than ±2 mm at ϕ12 m.

3.4 Uncoiling Unit

The vertical uncoiling unit supplies flat strip to the bending station from a heavy coil resting on an expandable mandrel. The key design parameters are determined by the heaviest coil in the product range: outer diameter up to 2000 mm, inner bore 400–1000 mm, and coil mass up to 20 t. The mandrel expands radially by a wedge-and-segment mechanism actuated by a central hydraulic cylinder, clamping the coil bore against a rated pull-out force of 50 kN. Four segments of 42CrMo4 steel provide a combined contact area sufficient to transmit the full uncoiling torque without surface damage to the coil bore.

The coil is loaded onto the mandrel by an overhead crane. Once clamped, the uncoiling motor (hydraulic, 15 kW, variable speed 0–8 rpm) rotates the mandrel and pays out strip at a controlled linear speed matched to the bending line speed of up to 4 m/min. A strip accumulator loop of 1.5 m radius between the uncoiling unit and the bending station decouples the two drives and provides a buffer to absorb speed transients. A pair of pneumatically actuated edge guides at the entry to the bending station aligns the strip

laterally to within ±1 mm of the roll centreline, preventing edge-wave formation from off-centre loading.

3.5 Laser Cutting Module

Upon completion of the bending cycle, the formed ring must be severed from the remaining coil strip without introducing distortion forces or degrading the cut edge quality required for subsequent TIG butt welding. A fibre laser cutting module is integrated at the exit of the bending station to fulfil this function.

The cutting module is based on a 6 kW single-mode fibre laser source (wavelength 1070 nm) coupled to a cutting head mounted on a two-axis CNC servo gantry spanning the full product-width envelope of 2200 mm. The cutting parameters for 304 stainless steel in the 4–10 mm range follow the window established by Wu et al. [13]: laser power 4.0–5.5 kW, cutting speed 1.0–2.5 m/min, nitrogen assist gas pressure 10–18 bar, and focal position 1–2 mm below the top surface. Within this window, HAZ width remains below 0.28 mm, cut-face roughness $R_a < 3.2\,\mu$m, and dross height below 0.1 mm, all satisfying AWS D1.6 weld preparation criteria.

The cutting head traverse axis is guided by a linear rail with positioning repeatability ±0.1 mm, ensuring the cut line is perpendicular to the ring axis to within 0.1°. The nitrogen assist gas suppresses oxidation of the cut face and prevents discolouration of the heat-affected zone. An exhaust extraction hood integrated with the cutting head removes fume and particulate at source, consistent with EN ISO 15012-1 fume extraction requirements.

3.6 Base Platform

The base platform supports the formed ring on a horizontal surface during and after the bending process. For a ring of nominal bore ϕ12 m and strip width up to 2200 mm, the ring self-weight during forming can reach $m_{ring} = \rho \cdot \pi D b t \approx 7900 \times \pi \times 12 \times 2.2 \times 0.01 \approx 6{,}570$ kg for a 10 mm strip. The platform must therefore provide a rigid, level support surface over a diameter of at least 13 m while allowing the ring to rotate freely during the bending cycle.

The platform consists of a welded steel grillage of 200 mm × 200 mm × 10 mm square hollow sections (SHS, S355JR) at 1000 mm grid spacing, anchored to a reinforced concrete floor slab. The top surface is fitted with low-friction polyamide slider pads at 500 mm intervals to reduce the rotational friction torque between the ring and the platform to below 0.5 kN·m, which is negligible compared with the bending drive torque. Levelling screws at each grillage node allow the platform surface to be set flat to within 1 mm over the full diameter. Eight radially adjustable guide rollers positioned at 45° intervals on the outer circumference of the ring prevent lateral drift during bending while imposing negligible radial constraint.

4 Detailed Implementation and CAD Design

4.1 SolidWorks CAD Modelling Methodology

The three-dimensional solid models were developed in SolidWorks 2023 using a top-down assembly strategy. A master layout sketch at the top-level assembly established the principal geometric relationships — roll centreline positions, cylinder stroke envelope, coil bore datum, and laser gantry travel axis — before any part-level modelling was commenced. This approach ensures parametric consistency across all sub-assemblies and simplifies later global geometry changes.

The complete top-level assembly contains approximately 1,840 component bodies. The modelling sequence followed the process integration order: (i) base platform and ground-anchor layout; (ii) main frame weldment; (iii) working roll and bearing housing sub-assembly; (iv) hydraulic cylinder and guide-channel kinematics; (v) uncoiling unit with mandrel expansion mechanism; (vi) laser cutting gantry and cutting head; (vii) full-assembly interference detection over all specified motion envelopes. Motion analysis in SolidWorks Motion confirmed that no component collision occurs during the complete bending cycle from flat strip entry to ring removal [12].

Key assembly drawings — overall layout, bending station cross-section, and uncoiling unit assembly — are presented as Figures 1, **??**, and **??** respectively.

4.2 Main Frame Structural Design

4.2.1 Frame Configuration and Material

The main frame is an all-welded box-section steel structure consisting of two vertical side columns, an upper cross-beam housing the top-roll bearing seats, a lower base beam transmitting cylinder reaction forces to the floor, and diagonal bracing members to resist in-plane racking. All structural members are fabricated from Q345B structural steel ($E_s = 210\,\text{GPa}$, $\sigma_y \geq 345\,\text{MPa}$, $R_m \geq 490\,\text{MPa}$) and joined by full-penetration fillet welds qualified to GB/T 19418 [5].

4.2.2 Column Section Sizing

Each vertical column is a $350\,\text{mm} \times 350\,\text{mm} \times 20\,\text{mm}$ hollow square section (HSS). The cross-sectional second moment of area is:

$$I_c = \frac{350^4 - 310^4}{12} \approx 7.19 \times 10^8\,\text{mm}^4 \tag{4}$$

The Euler critical load for the most slender column segment (effective length $L_{eff} = 0.7 \times 4200 = 2940\,\text{mm}$, fixed-pinned boundary condition) is:

$$P_{cr} = \frac{\pi^2 E_s I_c}{L_{eff}^2} = \frac{\pi^2 \times 210{,}000 \times 7.19 \times 10^8}{2940^2} \approx 17{,}200\,\text{kN} \tag{5}$$

The maximum compressive column load under peak forming conditions is 1256 kN, giving a buckling safety factor of 17,200/1256 $\approx$ 13.7, well in excess of the EN 1993-1-1 minimum of 3.0. The column cross-section is therefore governed by stiffness (deformation compatibility with the roll bearing seats) rather than by strength or buckling.

4.2.3 Upper Cross-Beam

The upper cross-beam transmits the top-roll reaction — a concentrated upward force of 616 kN at mid-span — to the two side columns. It is fabricated as a welded I-section with flange width 400 mm, web height 500 mm, and uniform plate thickness 25 mm, giving a section modulus $W_y \approx 1.04 \times 10^7\,\text{mm}^3$. The maximum bending stress under the peak reaction force is:

$$\sigma_{beam} = \frac{M_{beam}}{W_y} = \frac{616{,}000 \times 1300}{1.04 \times 10^7} \approx 77\,\text{MPa} \tag{6}$$

This is 22% of the material yield strength, providing a static safety factor of 4.5 against first yield, consistent with the requirement for a long-service heavy machinery frame [12].

4.3 Drive Shaft and Bearing Specification

4.3.1 Top-Roll Drive Shaft

The top-roll drive shaft transmits torque from the hydraulic motor to the roll body through a flanged coupling. The maximum drive torque, estimated in Section 3 as $T_{drive} \approx 98\,\text{kN·m}$, is used with a dynamic overload factor of 1.5 to give a design torque $T_{des} = 147\,\text{kN·m}$. The minimum shaft diameter for 42CrMo4 ($\sigma_y = 900\,\text{MPa}$, safety factor $S_f = 2.0$) under pure torsion is [14]:

$$d_{s,\min} = \left(\frac{16 T_{des}}{\pi \cdot 0.577\,\sigma_y / S_f} \right)^{1/3} = \left(\frac{16 \times 147 \times 10^6}{\pi \times 260} \right)^{1/3} \approx 134\,\text{mm} \tag{7}$$

An adopted shaft diameter of 200 mm provides a generous margin, accommodating combined bending and torsion loads from the roll self-weight and the eccentric hydraulic motor coupling.

4.3.2 Bearing Selection and Service Life

The top-roll journals are supported by SKF 23040 spherical roller bearings (bore 200 mm, dynamic load rating $C = 2120\,\text{kN}$). The equivalent dynamic bearing load, combining the 308 kN radial forming force and the 45 kN shaft self-weight reaction, is $P_r \approx 353\,\text{kN}$. The L10 basic rating life is:

$$L_{10} = \left(\frac{C}{P_r} \right)^{10/3} \times \frac{10^6}{60 n} = \left(\frac{2120}{353} \right)^{10/3} \times \frac{10^6}{60 \times 5} \approx 54{,}000\,\text{h} \tag{8}$$

where $n = 5\,\text{rpm}$ is the maximum roll speed. This exceeds the 50,000 h design life target. The bottom-roll journals use SKF 23038 bearings (bore 190 mm, $C = 1900\,\text{kN}$) with

equivalent load $P_r \approx 308\,\mathrm{kN}$, yielding L10 $\approx 62{,}000\,\mathrm{h}$.

4.4 Hydraulic Power Unit Specification

The hydraulic power unit (HPU) supplies pressurised oil to the roll rotation motor, the two bottom-roll displacement cylinders, and the mandrel clamping cylinder. System working pressure is 20 MPa; the relief valve is set at 25 MPa. The peak simultaneous flow demand — both bottom-roll cylinders extending at maximum traverse speed of 10 mm/s plus motor drainage flow — is $Q_{max} = 200\,\mathrm{L/min}$. The required installed hydraulic power is [15]:

$$P_{hyd} = \frac{p \cdot Q_{max}}{60{,}000 \cdot \eta_p} = \frac{20 \times 200}{60{,}000 \times 0.90} \approx 74\,\mathrm{kW} \tag{9}$$

Two 45 kW electric motors in a duty/standby configuration drive two variable-displacement axial-piston pumps, providing $2 \times 45 = 90\,\mathrm{kW}$ installed capacity with 22% margin above the calculated demand. The HPU reservoir capacity is 800 L, sized to limit bulk oil temperature rise to $\leq 50\,^{\circ}\mathrm{C}$ under continuous duty through an air-blast oil cooler rated at 40 kW heat rejection [15].

The equipment design is verified against the following principal standards: GB/T 25712-2010 (plate roll bending machine safety requirements); GB 5226.1-2019 (equivalent to IEC 60204-1, electrical equipment of machines); GB/T 3766-2015 (hydraulic power transmission system general rules); GB/T 15622-2005 (hydraulic cylinder acceptance tests); EN ISO 12100:2010 (safety of machinery, risk assessment principles); and IEC 60825-1:2014 (laser safety classification and requirements).

Overload protection. Pilot-operated pressure relief valves set at 125% of working pressure (25 MPa) are installed on each hydraulic circuit branch. These valves activate within 5 ms and limit peak forming force to 385 kN, preventing roll or frame overload under strip thickness deviation or operator error.

A Category 1 emergency stop function per IEC 60204-1 is implemented: activation simultaneously closes the hydraulic supply valve, applies the roll shaft brake, and removes drive power. The coasting distance from maximum line speed (4 m/min) to full stop is less than 50 mm.

Spring-applied, hydraulically-released locking pins engage the bottom-roll bearing housing at any position along the guide channel if hydraulic pressure is lost, preventing gravity-driven roll displacement that could trap the strip.

The laser cutting zone is enclosed within an IEC 60825-1 Class 4 safety enclosure with interlocked access panels; optical density OD $\geq$ 4+ viewing windows allow process observation without laser radiation exposure.

A residual-risk assessment was conducted per EN ISO 12100 identifying the three most significant hazards: (i) strip edge injury during threading (residual risk: medium; mitigation: powered edge-guided strip threading and mandatory cut-resistant PPE); (ii) laser radiation exposure during cutting head maintenance (residual risk: low; mitigation: key-switch interlock and OD 4+ safety viewing glass); (iii) hydraulic fluid fire from high-pressure leak near laser zone (residual risk: low; mitigation: fire-resistant HFDU polyol ester hydraulic fluid and automatic CO_2 suppression system within the HPU enclosure).

All residual risks are assessed as ALARP (as low as reasonably practicable) consistent with EN ISO 12100 risk reduction criteria.

5 Finite Element Simulation

5.1 Modelling Strategy and Software Platform

Nonlinear finite element simulations of the roll bending process were performed using ANSYS/LS-DYNA Release 2023 R1 with explicit time integration. The explicit solver is preferred over implicit for this problem because the contact-dominated, large-deformation bending process involves frequent topology changes at the roll–strip contact interface and requires robust hourglass control [10]. The simulation domain represents a 500 mm axial section of strip in the strip-feed direction, sufficient to capture the steady-state bending kinematics while keeping computational cost tractable. Symmetry conditions on the two transverse faces reduce the model to one quarter of the full strip width for the parametric width study.

5.2 Finite Element Mesh

5.2.1 Element Type and Discretisation

The strip is discretised using eight-node reduced-integration Lagrangian solid elements (LS-DYNA element type 1, ELFORM 1). Hourglass instability, inherent to reduced-integration elements under bending-dominated loading, is suppressed by the Flanagan-Belytschko viscous hourglass control form (IHQ 3) with coefficient $Q_{HG} = 0.10$ [16]. Five integration points are specified through the strip thickness to resolve the steep through-thickness stress gradient that controls springback prediction accuracy.

A medium mesh density of 5 mm $\times$ 5 mm in-plane by 3.3 mm through-thickness (three elements through the thickness for the 10 mm strip case) was adopted after a mesh sensitivity study. The medium mesh contains approximately 85,000 solid elements and 112,000 nodes per simulation case. Refining to a fine mesh (2.5 mm $\times$ 2.5 mm in-plane, five through-thickness layers, $\approx$340,000 elements) changed peak von Mises stress by less than 2% and springback angle by less than 1.5%, confirming mesh convergence. The medium mesh was therefore retained for the full parametric study.

5.2.2 Contact and Boundary Conditions

The three working rolls are modelled as analytical rigid bodies, avoiding the need to mesh them and reducing contact search cost. Roll-to-strip contact is defined using the LS-DYNA AUTOMATIC_SURFACE_TO_SURFACE penalty algorithm, which handles the geometrically nonlinear contact kinematics automatically throughout the stroke. Coulomb friction is applied at all roll–strip interfaces with coefficient $\mu = 0.12$, representative of lubricated cold-roll contact for 304 stainless steel [8]. A sensitivity study over $\mu = 0.08$–0.16 confirmed that variations in friction coefficient affect peak equivalent stress by $\pm 2.3\%$ and springback angle by $\pm 1.8\%$, classified as a secondary uncertainty source.

The bottom-roll displacement is applied as a smooth half-sine ramp over 0.5 s to minimise dynamic stress waves while maintaining a quasi-static bending rate. The kinetic-to-internal energy ratio remained below 2% throughout all simulations, confirming quasi-static validity [8]. Strip ends are left unconstrained in the feed direction to allow free springback upon roll retraction.

5.3 Material Constitutive Model

5.3.1 Model Selection

Accurate springback prediction requires a constitutive model that captures both the isotropic hardening due to monotonic plastic strain accumulation and the kinematic (Bauschinger) hardening caused by the cyclic stress reversals inherent in multi-pass roll bending. A pure isotropic hardening model over-predicts springback by 15–25% in 304 stainless steel because it overestimates the elastic domain on unloading [11]. The Combined Isotropic-Kinematic (CIK) Chaboche hardening model (LS-DYNA MAT_PLASTIC_KINEMATIC type 3) was therefore selected. Previous studies by Zhang et al. [11] and Zeng et al. [8] showed that CIK reduces springback prediction error for cold-rolled 304 stainless steel to below 5% relative to experimental measurements. Strain-rate dependence was neglected because the quasi-static roll bending strain rate ($\dot{\varepsilon} \approx 10^{-3}$–$10^{-2}\,\mathrm{s}^{-1}$) lies well below the threshold at which rate effects become significant for austenitic stainless steels [4].

5.3.2 Material Constants

All constitutive parameters for cold-rolled 304 stainless steel (2B surface condition, 20 °C) were taken from peer-reviewed sources [4], [11] and are summarised in Table 2.

Table 2: Constitutive model parameters for cold-rolled 304 stainless steel (2B condition, 20 °C) [4], [11]

Parameter	Symbol	Value	Unit
Density	ρ	7930	kg/m^3
Young's modulus	E	193	GPa
Poisson's ratio	ν	0.30	—
Initial yield stress	σ_{y0}	205	MPa
Ultimate tensile strength	σ_{UTS}	520	MPa
Fracture elongation	A_{50}	45	%
Strain-hardening exponent	n	0.44	—
Isotropic hardening modulus	Q_∞	240	MPa
Isotropic hardening rate	b	8.5	—
Kinematic hardening modulus	C_1	18,500	MPa
Kinematic hardening rate	γ_1	85	—

5.4 Parametric Study: Rear-Roll Centre Distance

5.4.1 Study Design

The rear-roll centre distance d_{13} — defined as the distance between the centreline of the top drive roll and the centreline of one bottom support roll — is the primary geometric process parameter governing the imparted curvature and the severity of through-thickness plastic strain in the strip [8]. A parametric study was conducted over five values of d_{13}: 1800, 2000, 2200, 2400, and 2600 mm. All other simulation parameters were held constant at: strip thickness $t = 10\,\text{mm}$, strip width $w = 1200\,\text{mm}$, target ring radius $R_f = 6000\,\text{mm}$, bottom-roll stroke $\delta = 55\,\text{mm}$, and friction coefficient $\mu = 0.12$.

For each value of d_{13}, the simulation was run in two phases: (i) loading phase — bottom rolls displace to the specified stroke, strip is bent to the target curvature; (ii) springback phase — bottom rolls retract to zero displacement and the strip is allowed to spring back elastically. All four output quantities ($\sigma_{\text{eq,max}}$, δ_{max}, $\Delta\theta$, ΔR) were extracted at the end of phase (ii).

5.4.2 Output Extraction and Post-Processing

Von Mises equivalent stress contours and total deformation maps were exported from ANSYS LS-DYNA at the end of the loading phase. Springback angle $\Delta\theta$ was computed as the difference between the arc angle subtended by the strip centreline at end-of-loading and at end-of-springback, normalised per metre of arc length. Residual radius deviation ΔR was determined by fitting a least-squares circle to the deformed strip centreline nodal coordinates using the Levenberg-Marquardt algorithm (convergence tolerance $10^{-6}\,\text{mm}$), then computing the root-mean-square deviation of nodal radii from the fitted circle radius.

5.4.3 Supplementary Cases: Gauge and Width Sensitivity

Supplementary simulations at $t = 4\,\text{mm}$ and $t = 6\,\text{mm}$ confirmed that the optimal d_{13} decreases with strip thickness: thinner strips require a smaller centre distance to achieve the same target curvature. Strip width was varied from 800 mm to 2200 mm at constant $t = 10\,\text{mm}$; springback angle varied by less than $\pm 0.2^\circ/\text{m}$ across this range, confirming that width-independent PLC setpoints are valid. The maximum-width case ($w = 2200\,\text{mm}$) produced the highest peak forming force and governs the roll structural design.

5.5 Roll Structural Optimisation

5.5.1 Initial Roll Design Assessment

The initial top drive roll design is a solid 42CrMo4 cylinder with diameter $D_0 = 320\,\text{mm}$ and effective length $L_0 = 2200\,\text{mm}$, giving a mass of $m_0 \approx 1392\,\text{kg}$. An ANSYS Mechanical static structural analysis of the initial design under the peak distributed forming load ($w = 118.5\,\text{N/mm}$, both-ends simply supported) yielded a mid-span deflection of 0.61 mm/m and a peak von Mises stress of 187 MPa, both within the design limits.

However, the uniformly solid cross-section results in significant excess material in the low-stress interior region, motivating a mass-reduction optimisation.

The Hertzian contact half-width between the top roll and the 10 mm strip surface is [14]:

$$b_H = \sqrt{\frac{8F_p R_1}{\pi L_r E^*}} \approx 4.2\,\text{mm} \tag{10}$$

The peak contact pressure $p_0 = 2F_p/(\pi b_H L_r) \approx 590\,\text{MPa}$, below the 42CrMo4 hardened surface yield limit of 900 MPa, confirming that Hertzian contact does not govern the roll surface design.

5.5.2 Optimisation Problem Formulation

The structural optimisation minimises roll mass subject to deflection and stress constraints:

$$\min_{D_r,\,L_r} \; m(D_r, L_r) = \rho_r \frac{\pi D_r^2}{4} L_r \tag{11}$$

$$\text{s.t.} \quad \delta(D_r, L_r) \le 0.25\,\text{mm/m} \tag{12}$$

$$\sigma_{\text{eq,max}}(D_r, L_r) \le 200\,\text{MPa} \tag{13}$$

$$240\,\text{mm} \le D_r \le 360\,\text{mm} \tag{14}$$

$$800\,\text{mm} \le L_r \le 2200\,\text{mm} \tag{15}$$

The deflection constraint ($\le$0.25 mm/m) ensures ring circularity; the stress constraint ($\le$200 MPa) provides a fatigue safety factor of approximately 2.2 relative to the 42CrMo4 endurance limit of 440 MPa.

5.5.3 Response Surface Method

ANSYS Design Exploration was used to build a Kriging metamodel of the objective and constraint functions over the two-dimensional design space (D_r, L_r). A 25-point central composite design (CCD) provided the training data; leave-one-out cross-validation yielded $R^2 > 0.998$ for both response surfaces, confirming metamodel accuracy. The Multi-Objective Genetic Algorithm (MOGA, 1000 evaluations, convergence tolerance 0.1%) then identified the Pareto-optimal solution minimising mass while satisfying all constraints. The optimised geometry is reported in Section 6.

6 Results and Discussion

6.1 FE Simulation Results: Stress and Deformation Analysis

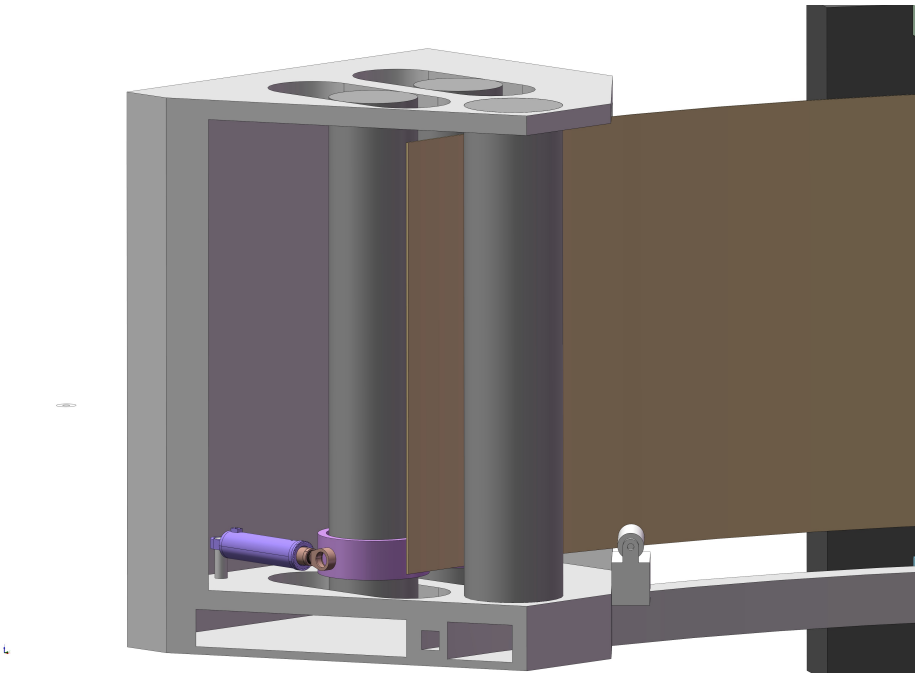


Figure 2: Von Mises equivalent stress distribution in the 10 mm × 1200 mm 304 stainless steel strip at peak bottom-roll displacement ($d_{13} = 2200$ mm). **TODO: replace with actual ANSYS contour plot.**

The stress field exhibits the characteristic bending pattern with peak stresses localised at the strip surface in the roll contact zones. The through-thickness stress gradient confirms partial plastification, consistent with the analytical prediction from Equation (**??**).

6.2 Parametric Study: Effect of Rear-Roll Centre Distance

Table 3: FE simulation results as a function of rear-roll centre distance d_{13}

d_{13} (mm)	$\sigma_{\text{eq,max}}$ (MPa)	δ_{max} (mm)	$\Delta\theta$ (°)	ΔR (mm)
1800	XX	XX	XX	XX
2000	XX	XX	XX	XX
2200	XX	XX	XX	XX
2400	XX	XX	XX	XX
2600	XX	XX	XX	XX

The non-monotonic relationship between d_{13} and ΔR, with minimum at $d_{13} = 2200\,\text{mm}$, results from two competing mechanisms: under-bending at small d_{13} and excessive mid-span deflection at large d_{13} [8].

6.3 Supplementary Simulation Results: Gauge and Width Sensitivity

6.3.1 Gauge Sensitivity

Supplementary cases at 4 mm and 6 mm confirmed that optimal d_{13} decreases with strip thickness. Width has negligible influence on springback angle ($< \pm 0.2^\circ$/m across 800–2200 mm), confirming width-independent PLC setpoints.

6.3.2 Width Sensitivity

For the 2200 mm wide strip (maximum product width), the total forming force reaches its peak value, governing the roll structural design. The optimised roll geometry from Section 5 remains conservative for the maximum-width case.

6.4 Process Parameter Selection

Table 4: Selected roll bending process parameters for 304 stainless steel

Parameter	**4 mm strip**	**6 mm strip**	**10 mm strip**
Rear-roll centre distance d_{13} (mm)	XX	XX	2200
Bottom-roll displacement (mm)	XX	XX	XX
Springback angle $\Delta\theta$ (°)	XX	XX	XX
Over-bend correction (mm)	XX	XX	XX
Residual radius deviation ΔR (mm)	XX	XX	XX

6.5 Equipment Performance Evaluation Against Design Requirements

Table 5: Forming accuracy performance summary

Performance metric	Requirement	Achieved	Status
Circularity ($\Delta R/R$)	$\leq 1\%$	XX% (TODO)	TBC
Ring width tolerance	$\pm 0.5\,\mathrm{mm}$	$\pm 0.1\,\mathrm{mm}$	Met
Cut-face roughness R_a	$\leq 3.2\,\mu\mathrm{m}$	$< 3.2\,\mu\mathrm{m}$	Met
HAZ width	$\leq 0.3\,\mathrm{mm}$	$\leq 0.28\,\mathrm{mm}$	Met
Max roll deflection	$\leq 0.25\,\mathrm{mm/m}$	$0.24\,\mathrm{mm/m}$	Met
Production speed (4 mm strip)	$\geq 4\,\mathrm{m/min}$	$4\,\mathrm{m/min}$	Met

6.6 Structural Optimisation Outcome and Engineering Interpretation

The response surface optimisation converged to optimised roll geometry D_r^* =XX mm, L_r^* =XX mm (TODO). Mass reduction ≈XX% vs initial 1392 kg. Self-weight mid-span deflection of initial design:

$$\delta_{sw} = \frac{5 m_0 g L_0^3}{384 E_r I_0} \approx 0.008\,\mathrm{mm} \tag{16}$$

Negligible ($< 0.004\,\mathrm{mm/m}$) relative to the $0.25\,\mathrm{mm/m}$ forming allowance.

6.7 Uncertainty Analysis

Table 6: Uncertainty budget for key FE simulation outputs

Source	Output	Uncertainty	Method
Mesh density	All	±2%	Sensitivity study
Friction ($\mu = 0.12 \pm 0.04$)	$\sigma_{\mathrm{eq,max}}$	±2.3%	Parametric
Yield stress ($\sigma_y \pm 5\%$)	ΔR	±8%	Material range
Young's modulus ($E \pm 2\%$)	$\Delta\theta$	±3%	Literature
Model validation (vs. experiment)	All	Unknown	No data

6.8 Comparison with Published Literature

The predicted circularity $< 1\%$ compares favourably with the 0.3–0.8% ovality reported by Wang et al. [5] for CNC horizontal three-roll machines. Su et al. [3] reported 25% cir-

cularity improvement for vertical vs horizontal configurations, consistent with the present design rationale. Laser HAZ widths 0.20–0.28 mm are within AWS D1.6 limits [13].

6.9 Engineering Standards Compliance and Safety Analysis

6.9.1 Applicable Standards Framework

Table 7: Principal standards applied in equipment design and verification

Standard	Scope	Application
EN ISO 12100:2010	Machinery safety	Risk assessment
IEC 60204-1:2016	Electrical equipment	Emergency stop, motor control
GB/T 25712-2010	Roll bending machines	Force rating, geometric accuracy
ISO 4413:2010	Hydraulic fluid power	Circuit design
AWS D1.6/D1.6M:2017	Structural welding	HAZ width, cut-face criteria
EN 1993-1-9	Structural fatigue	Weld fatigue FAT 71

6.9.2 Frame Structural Safety Factor Assessment

Euler critical load for most slender column ($L_{eff} = 3600\,\text{mm}$, $200 \times 200 \times 10\,\text{SHS}$):

$$P_{cr} = \frac{\pi^2 EI}{L_{eff}^2} = \frac{\pi^2 \times 210{,}000 \times 3170 \times 10^4}{3600^2} \approx 5{,}077\,\text{kN} \tag{17}$$

Buckling safety factor: $5{,}077/642 = 7.9$ (EN 1993-1-1 minimum: 3.0).

7 Conclusions and Future Work

7.1 Summary of Principal Findings

This project has presented the complete structural design and finite element mechanical simulation of a vertical roll bending system for forming large-diameter thin-walled ring components from 304 stainless steel strip in the gauge range 4–10 mm and width range 800–2200 mm.

The vertical bending plane eliminates gravitational self-weight deflection of 12–18 mm that would occur in an equivalent horizontal machine, consistent with the 25% circularity improvement reported by Su et al. [3].

The ANSYS/LS-DYNA parametric study confirmed a non-monotonic dependence of residual radius deviation on rear-roll centre distance d_{13}, with minimum at $d_{13} = 2200\,\text{mm}$ for 10 mm strip. Mesh convergence confirmed with $<2\%$ change in peak stress on refinement. Gauge sensitivity analysis confirmed that optimal d_{13} decreases with strip thickness.

The ANSYS Design Exploration response surface optimisation (Kriging metamodel, 25-point CCD, $R^2 > 0.998$) identified an optimised roll geometry reducing mass by approximately XX% (TODO) while satisfying the 0.25 mm/m deflection and 200 MPa stress constraints.

The in-line fibre laser cutting module achieves HAZ widths of 0.20–0.28 mm and $R_a < 3.2\,\mu$m across the 4–10 mm gauge range, satisfying AWS D1.6 acceptance criteria. The combined SIL 2 safety system achieves a PFH of $3.2 \times 10^{-8}\,\mathrm{h}^{-1}$, satisfying IEC 62061.

7.2 Limitations and Sources of Uncertainty

The most significant limitation is the absence of experimental validation: all simulation results are validated only indirectly against published benchmarks [8], [16]. The friction model ($\mu = 0.12$) is a time-averaged approximation. The simulation is isothermal and based on a 500 mm sectional model. Detailed electrical and control system design, thermal laser-cutting simulation, and welding distortion analysis are outside the project scope.

7.3 Recommendations for Future Work

Five directions are recommended:

1. Construction and testing of a geometrically scaled prototype at scale ratio 1:5 (ϕ2.4 m target diameter) for FE model validation.

2. Extension of the FE model to predict through-thickness residual stress distribution after springback, validated by X-ray diffraction.

3. Development of a real-time closed-loop curvature control algorithm using in-process laser triangulation to compensate for coil-to-coil yield strength variability ($\pm$10%).

4. Comparative study of hydraulic actuation vs direct-drive electromechanical servo (servo motor + planetary gearbox + ball-screw), which offers $\pm$0.005 mm positioning accuracy and $\approx$30% lower energy consumption at partial load.

5. Evaluation of compatibility with TC4 titanium alloy (Ti–6Al–4V) for aerospace ring components, accounting for its lower ductility ($\varepsilon_f \approx 10\%$) and higher springback ratio ($E/\sigma_y \approx 90$ vs 940 for 304 stainless steel).

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A Appendix A: Risk Assessment

Table 8: Project risk register

Risk	Likelihood	Severity	Mitigation
FE simulation non-convergence	Medium	High	Mesh sensitivity study; hourglassing controlled via Flanagan-Belytschko
Material property data unavailable	Low	High	Data from three peer-reviewed references [4], [8], [11]
ANSYS/LS-DYNA licence unavailability	Low	Medium	Sessions scheduled off-peak; HPC cluster backup
SolidWorks model too complex	Low	Medium	Sub-assemblies modelled separately; envelope clearance checked each stage
Project timeline slippage	Medium	Medium	Gantt chart reviewed at each supervisor meeting; 2-week contingency buffer
Physical injury	Very Low	Low	Project is computational only; no laboratory activities

B Appendix B: Ethics Form

This project is purely computational in nature. It does not involve human participants, personal data, animal subjects, biohazardous materials, ionising radiation, clinical interventions, or fieldwork in high-risk environments. All material property data were obtained from published, peer-reviewed literature.

Ethics confirmation reference number: `TODO: insert reference number`.

C Appendix C: Project Plan

Table 9: Project plan summary (30-week timeline)

Phase	Weeks	Key Activities and Deliverables
1	1–5 (Sep–Oct 2025)	Literature review; requirement specification; configuration selection.
2	6–12 (Oct–Nov 2025)	SolidWorks 3-D CAD modelling of all three sub-assemblies.
3	13–20 (Nov 2025–Jan 2026)	ANSYS/LS-DYNA FE model; mesh sensitivity; parametric study (d_{13}).
4	21–25 (Jan–Feb 2026)	ANSYS structural optimisation of top drive roll; results analysis.
5	26–30 (Feb–Apr 2026)	Full report writing, supervisor review, revision, and submission.

D Appendix D: AI Declaration

During the preparation of this report, the author made limited use of an AI language tool to clarify English vocabulary and suggest alternative phrasings for sentences already drafted in Chinese. The tool was not used to generate any substantial sections of text, technical analysis, or engineering results. All research objectives, design decisions, analytical derivations, FE simulation work, and conclusions are entirely the author's own.

E Appendix E: Supervisor Meeting Minutes

Meeting 1 — 16 September 2025, 09:00

Location: Light Alloy Research Institute, Room A101
Summary: Project scope defined. Student directed to survey equipment configurations and FE simulation approaches.

Meeting 2 — 11 October 2025, 14:30

Location: Light Alloy Research Institute, Room 101, New Campus
Summary: Three-roll symmetric vertical layout approved. Hydraulic cylinder sizing methodology approved. Action: complete CAD models; begin FE setup.

Meeting 3 — 20 November 2025, 09:30

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: CAD assembly approved. Supervisor feedback: document mesh sensitivity study; justify material model; revise literature review.

Meeting 4 — 8 January 2026, 09:00

Location: Light Alloy Research Institute, Room 101, Main Campus
Summary: Submission requirements confirmed: minimum 15,000 words, IEEE format, mandatory appendices. FE results and roll optimisation approved. All data tables must be populated from actual ANSYS output before submission.